Rocio Perales Valdes

Atlanta, Georgia • rocio.perales@gatech.edu • rociopv.com

Education

Georgia Institute of Technology | BS Computer Science | GPA: 3.92/4.0 May 2028
Minor: Law, Science & Technology
Relevant Coursework: Applied Combinatorics; Computational & Model Algorithms; DSA; Linear Algebra; Discrete Math

Skills

Programming: Python, Java, C++, JavaScript, HTML/CSS
AI/ML: PyTorch, JAX, TensorFlow, Sparse Autoencoders (SAEs), Slurm, VLAs, ROS2, PID Control
Languages: English, Spanish, French (conversational)

Professional & Research Experience

Undergraduate Researcher | PAIR Lab – Georgia Tech Nov. 2025 – Present
- Leading interpretability goal-oriented SAE-based work on VLAs ($\pi_{0.5}$). Advised by Stepan Shabalin and Prof. Animesh Garg.
- Assisted in ICRA paper on Progress Reward Model training achieving a 10-15% performance jump on precise tasks.

Summer Research Fellow | Institute for AI Policy and Strategy (IAPS) Jun. 2026 – Aug. 2026
- Synthesized a 269-page Great American AI Act discussion draft into a 2-page decision brief for the **Policy Team**
- Authored a public explainer and policy recommendations on the FCC's proposed ban on foreign-made advanced robotics, securing review from 5 policy experts through direct outreach
- Conducted internal robotics threat modeling for the **Frontier Security** team, including BOTEC calculations on drone attacks and interviews with 6 stakeholders across industry, military, academia, and think tanks

ML Solutions Intern | ZeruDrones Jun. 2025 – Aug. 2025
- Trained **YOLO models** on AWS GPU instances and integrated **autoencoders** for powerline anomaly detection

Solo Developer | Personalized Word Count for Google Docs Jun. 2024 – July 2024
- Google Docs extension allowing for tailored word counts with 3K+ downloads

Leadership & Campus Involvement

Co-Director | AI Safety Initiative (AISI.dev) Jan. 2026 – Present
- Lead a 15-person exec organization with $30K+ of external funding and set overall strategy
- Created a Policy Fellowship curriculum run as a course seminar with Georgia Tech's OMSCS program
- Led **deepfake governance research** under **Dr. Margaret Kosal**

Chair | AI & Student Body Subcommittee – Georgia Tech Jan. 2026 – Present
- Selected by Provost Raheem Beyah to chair subcommittee for Georgia Tech's institutional AI strategy
- Designed and deployed a campus-wide AI usage survey collecting **1K+ responses** without monetary incentives

Software Developer | RoboJackets – RoboCup SSL Aug. 2024 – Sept. 2025
- Led implementation on the multi-robot pathfinding system (C++/ROS2, real-time decision-making under uncertainty); placed **2nd in the technical challenge** and **4th overall** at RoboCup 2025 Intl

Awards & Recognition

Stamps Walk-On President's Scholarship | Georgia Institute of Technology March 2026
- Competitively selected for full-ride scholarship recognizing academic achievement and leadership

AI@ATL Hackathon Winner | Major League Hacking Oct. 2024
- Built a **synthetic medical dataset** and **prompt injection testing framework**; **1st place**, AI Safety track

Miscellaneous Recognition:
- AI Safety, Ethics, and Society Course, Center for AI Safety (Nov. 2025 – Feb. 2026)
- Spanish Translation Assistance, METR & Redwood Research OpenAI/HF Incident Investigation (Aug. 2026)

Conferences: Economic Futures Symposium (DC, Sept. 2025)*, ControlConf (Berkeley, Apr. 2026), OASIS (Berkeley, Jan. 2026)*, DCMC (DC, Apr. 2026)*, Verified: The Conference (DC, Sept. 2026)*†, GCP (Bay Area, Aug. 2026)*
*Invited attendee, closed-door †Helped organize

Personal interests: Blogging on AI futures (Substack: rociopv), music composition and production, running, soccer